

KARTHICK K

Full Stack Developer | Generative AI | Agentic AI

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PROFESSIONAL SUMMARY

Full Stack Developer with 3+ years of experience shipping production systems for government and healthcare platforms using React.js and Python Flask. Successfully cut API latency by 98%, led a UIDAI PII-protection audit to achieve 100% compliance, and maintained a 100,000-transaction voting platform during peak loads. Experienced in building Generative AI features in production, including a LangChain/RAG chatbot on a Chroma vector store and an intelligent interview assessment tool, backed by self-study in LangGraph and advanced prompt engineering.

TECHNICAL SKILLS

Frontend: ReactJS, TypeScript, JavaScript (ES6+), HTML5, CSS3, Ant Design

Backend: Python, Flask, RESTful API Design, Microservices, Swagger/OpenAPI

Generative & Agentic AI: LangChain, RAG pipelines, Chroma vector store, LangGraph, MCP, AI-driven automation

Data: MongoDB (indexing, aggregation), PostgreSQL, Redis

Security: JWT, RBAC, cryptographic encryption, UIDAI/Aadhaar PII compliance

Quality & Tooling: PyTest, Docker, Git, Bitbucket, CI-driven release workflows

PROFESSIONAL EXPERIENCE

AICARE Global Private Ltd - Full Stack Developer

Tamil Nadu, India | Dec 2022 - Present

- Decoupled a monolithic healthcare platform into Flask microservices, raising horizontal scalability by 40% and shortening deployment downtime.
- Tuned MongoDB indexing across 50+ endpoints, dropping query latency from 15s to under 300ms for high-volume patient records.
- Established a JWT + RBAC authentication layer protecting medical telemetry for 10,000+ practitioners under full regulatory compliance.
- Increased test coverage to 85%+ across core Flask services using PyTest, supporting weekly production releases.
- Published 25+ reusable ReactJS UI components across product modules, reducing code duplication and improving consistency.

KEY PROJECTS

Tamil Nadu Medical Council

Nov 2024 - Present

- Built an allocation engine (React + Flask) routing 500+ daily practitioner applications, lifting administrative throughput by 60%.
- Directed a UIDAI-mandated Aadhaar audit, reaching 100% PII-protection clearance across all checks.
- Onboarded 50,000+ biometric profiles via third-party fingerprint hardware APIs with zero mismatches.
- Built a LangChain and Chroma RAG chatbot indexing 5,000+ compliance documents, achieving a 95% retrieval accuracy rate.

Government Digital Election Platform

Dec 2023 - Sep 2024

- Architected a voting platform processing 100,000+ ballots in 24 hours with zero record loss via cryptographic encryption.
- Shipped real-time React/MongoDB dashboards delivering live vote tallies to stakeholders in under 1 second during a 100,000+ ballot election platform.
- Resolved critical L3 incidents during live polling while maintaining 100% server uptime under peak concurrency.

Enterprise Vendor Management SaaS

Aug 2023 - Jan 2024

- Launched a multi-tenant SaaS ecosystem with domain-driven isolation for 8+ enterprise clients.
- Improved read performance by 70% through MongoDB query optimisation and real-time sync.
- Developed an AI-driven interview automation system that processed 100+ candidate assessments and cut screening time by 50%.

EDUCATION

B.E., Electrical and Electronics Engineering - Adhiyamaan College of Engineering

2019 – 2023

CGPA: 8.5 / 10.0

CERTIFICATIONS

2026: AWS Cloud Practitioner Essentials - Amazon Web Services

2025: MongoDB Indexes - MongoDB University; React JS Frontend Development - Udemy

2022: Backend Development: Python - CSC, CISCO

AWARDS & RECOGNITION

- Certificate of Recognition, 2 Years of Excellence | AICARE Global Private Ltd (Jan 2025)
- Certificate of Appreciation, Election Support | Women Lawyers' Association (Sep 2024)
- Certificate of Appreciation, Election Support | Madras High Court Advocates' Association (Dec 2023, Apr 2026)